

THE SPHERICAL REGGE SCISSORS THEOREM: A SELF-CONTAINED PROOF THROUGH WEIGHT-TWO MOTIVES

ABSTRACT. We prove that every nondegenerate spherical tetrahedron is scissors congruent to each elementary Regge mate. This text contains only the proof of that statement and the background needed to read it. In particular, we derive the monomial Regge action on angle phases, rationalize the orientation double cover of the Gram-parameter space, explain the particular relative quadric motive used in the proof, compute its three weight pieces and its coproduct, and reduce the sole primitive obstruction to $H^1(U, \mathbb{Q}(2)) = 0$. We then spell out the specialization to the given tetrahedron and the final passage from Goncharov’s quadric invariant to a finite spherical equidecomposition. General motivic theory and alternative chain constructions are not developed beyond the precise standard results used in this argument.

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1. THE THEOREM AND THE LOGICAL ROUTE

1.1. Spherical scissors congruence. Let $S^3 \subset \mathbb{R}^4$ be the unit sphere. A spherical polytope is a finite union of geodesic simplices contained in open hemispheres. The spherical scissors group $\widehat{\mathcal{P}}(S^3)$ is generated by symbols $[P]$, one for each such polytope, subject to

$$[P] = [P_1] + \cdots + [P_m]$$

when the P_i have disjoint interiors and union P , and $[P] = [gP]$ for every spherical isometry g . Thus an equality in $\widehat{\mathcal{P}}(S^3)$ is the stable algebraic form of equidecomposability. A cancellation theorem recalled in Section 5 turns the equality proved below into an actual finite equidecomposition.

Label the vertices of a nondegenerate spherical tetrahedron by 0, 1, 2, 3, and select the opposite edge pair 01, 23. The elementary Regge operation fixes the two corresponding edge lengths and replaces the remaining four lengths $\ell_1, \dots, \ell_4$ by

$$\ell_i^R = \sigma_\ell - \ell_i, \quad \sigma_\ell = \frac{1}{2} \sum_{j=1}^4 \ell_j. \quad (1.1)$$

For a genuine mate, the same formula holds for the corresponding four dihedral angles, with their half-sum in place of σ_ℓ . This is the spherical Regge theorem of Akopyan–Izestiev [AI19]; we state the exact input in Theorem 2.3.

Theorem 1.1 (Spherical Regge scissors congruence). *Let $T \subset S^3$ be a non-degenerate spherical tetrahedron and let T^R be any nondegenerate elementary Regge mate. Then*

$$[T] = [T^R] \quad \text{in } \widehat{\mathcal{P}}(S^3).$$

Consequently T and T^R admit a finite spherical equidecomposition.

No algebraicity assumption is made: the angles and vertices may be transcendental.

1.2. What is proved and what is imported. The proof uses four established inputs. They are quoted in the precise forms needed later.

- (I1) **Regge geometry.** The same orthogonal linear involution acts on the four moving edge lengths and on the four moving dihedral angles, and Regge mates have the same spherical volume [AI19].
- (I2) **Motivic functoriality and localization.** Rational constructible motives over a smooth $\mathbb{Q}$ -scheme have pullback, tensor products, internal duals, Tate twists, localization triangles, and the projective-bundle formula. We use these operations only for complements of quadrics and their coordinate sections. Section 3 defines the resulting object explicitly and lists every formal property used.
- (I3) **Weight-two rigidity.** Motivic cohomology in weight two is birational in degrees at most one, and the corresponding field groups are unchanged by a purely transcendental extension [Gar23]. Borel’s theorem gives $\mathrm{gr}_2^\gamma K_3(\mathbb{Q}) \otimes \mathbb{Q} = 0$ [Bor77].
- (I4) **Goncharov’s comparison.** An oriented quadric with four ordered hyperplane faces defines a weight-two quadric-scissors class and a framed relative motive. Goncharov’s canonical map from primitive weight-two quadric-scissors classes to $\mathrm{gr}_2^\gamma K_3(\mathbb{C}) \otimes \mathbb{Q}$ is injective, and his spherical map to the quadric complex is injective in degree one [Gon99, Theorems 4.13 and 4.14].

Everything else—the parameter space, its rationality, the relative quadric calculation, the Regge coproduct identity, descent from the one nonsplit face, specialization, and the final scissors argument—is proved here.

1.3. A map of the argument. The proof has the following seven steps.

1. Replace each dihedral angle θ by its phase $u = e^{i\theta}$. The additive Regge formula then becomes the monomial substitution $u_i^R = \tau/u_i$.
2. Adjoin the orientation root $w = \sqrt{\det G}$ of the angle Gram matrix. Two elementary substitutions, $r = w - sx$ and $r_+ = w + sx$, rationalize this double cover. Every physical Regge pair is a complex point of one of the resulting rational opens U .
3. Associate to the universal Gram matrix and its mate two oriented quadric simplices over U . Their normalized relative motives have only Tate degrees 0, 1, 2. Five binary face sections split; the sixth is the common quadratic cover determined by the fixed phase p , whose sign summand is an Artin motive χ .
4. The middle coproduct of a dual quadric simplex is $\sum_e L_e \otimes A_e$, where L_e and A_e are its oriented length and angle cross-ratios. The Regge matrix is orthogonal, so the coproducts of the two mates agree. Trace from the fixed quadratic cover shows that this equality already holds over U .
5. A framed three-weight object with zero middle coproduct can differ from zero only by a direct endpoint extension. Here that obstruction is

$$\mathrm{Ext}^1(\mathbb{Q}_U(0), \mathbb{Q}_U(2)) = H^1(U, \mathbb{Q}(2)).$$

Weight-two birational invariance reduces this group to $\mathrm{gr}_2^\gamma K_3(\mathbb{Q}) \otimes \mathbb{Q} = 0$. Hence the universal framed defect vanishes.

6. Pull the identity back to the complex point representing the given tetrahedron. Goncharov's weight-two injectivity turns the motivic identity into equality of the associated primitive quadric-scissors classes, and then into equality in the reduced spherical scissors group.
7. The unreduced difference is a suspension of a spherical polygon. Regge volume equality makes that polygon have area zero; area classifies spherical planar scissors classes. Therefore the suspension is zero.

The next section begins at Step 1 and explains why the word “monomial” is literal rather than metaphorical.

2. FROM SPHERICAL ANGLES TO RATIONAL ORIENTATION CHARTS

This section replaces the trigonometric Regge transformation by an algebraic family. There are three steps. First, exponentiating the dihedral angles turns the linear Regge transformation into a monomial involution of an algebraic torus. Second, adjoining a square root of the Gram determinant retains the orientation of the associated quadric. Third, a change of variable rationalizes this double cover. The only angle phase which is lost in the rationalization is recovered by one explicit quadratic finite étale cover.

No motivic input is used in this section.

2.1. The spherical angle Gram matrix. Let $T \subset S^3$ be a nondegenerate spherical tetrahedron contained in an open hemisphere. Write its four facets as $F_0, \dots, F_3$, and let $n_i \in S^3 \subset \mathbf{R}^4$ be the outward unit normal to F_i . If F_i and F_j meet at interior dihedral angle $\theta_{ij} \in (0, \pi)$, then their outward normals meet at the supplementary angle. Hence

$$n_i \cdot n_j = -\cos \theta_{ij}. \quad (2.1)$$

The geometric edge at which these facets meet is indexed by the complementary pair of vertices. We use the facet-pair labels ij , since they are the natural labels for the Gram matrix. Complementing all six labels preserves opposite pairs and merely changes the edge-label convention.

Definition 2.1 (Angle Gram matrix). The angle Gram matrix of T is

$$G(T) = (g_{ij})_{0 \leq i, j \leq 3}, \quad g_{ii} = 1, \quad g_{ij} = -\cos \theta_{ij} \quad (i \neq j). \quad (2.2)$$

If N is the matrix with columns $n_0, \dots, n_3$, then $G = N^t N$. In particular, a nondegenerate spherical tetrahedron has

$$G > 0, \quad \det G = (\det N)^2 > 0. \quad (2.3)$$

The positivity is used only to locate the physical real locus. All algebraic constructions below allow the entries of G to be complex.

Fix the opposite Gram positions 01 and 23. In the cyclic order

$$(01, 02, 12, 23, 13, 03), \quad (2.4)$$

write the two fixed angles as θ_{01}, θ_{23} and the four moving angles as

$$\boldsymbol{\theta} = (\theta_{02}, \theta_{12}, \theta_{13}, \theta_{03})^t. \quad (2.5)$$

Put $\mathbf{1} = (1, 1, 1, 1)^t$, let $J_4 = \mathbf{1}\mathbf{1}^t$, and define

$$R = \frac{1}{2}J_4 - I_4. \quad (2.6)$$

If

$$S = \frac{1}{2}(\theta_{02} + \theta_{12} + \theta_{13} + \theta_{03}), \quad (2.7)$$

then

$$R\boldsymbol{\theta} = (S - \theta_{02}, S - \theta_{12}, S - \theta_{13}, S - \theta_{03})^t. \quad (2.8)$$

Lemma 2.2 (Elementary properties of the Regge matrix). *The matrix R is a symmetric orthogonal involution:*

$$R^t = R, \quad R^2 = I_4.$$

It is the identity on $\mathbf{Q1}$ and minus the identity on $\mathbf{1}^\perp$.

Proof. Since $J_4^2 = 4J_4$,

$$R^2 = \frac{1}{4}J_4^2 - J_4 + I_4 = I_4.$$

Symmetry is immediate. Also $R\mathbf{1} = \mathbf{1}$, whereas $J_4v = 0$ and hence $Rv = -v$ for $v \in \mathbf{1}^\perp$. $\square$

Theorem 2.3 (Geometric Regge input). *An elementary Regge move fixes θ_{01}, θ_{23} and replaces $\boldsymbol{\theta}$ by $R\boldsymbol{\theta}$. The same matrix acts on the corresponding four edge lengths, and the two tetrahedra have equal volume.*

Proof. This is the constant-curvature Regge theorem of Akopyan and Izestiev [AI19, Theorem 1]. Only the angle statement is needed in the present section. $\square$

2.2. Phase coordinates and the meaning of “monomial”. For a physical tetrahedron define its six angle phases by

$$\begin{aligned} p &= e^{i\theta_{01}}, & a &= e^{i\theta_{02}}, & b &= e^{i\theta_{12}}, & \tau &= e^{iS}, \\ q &= e^{i\theta_{23}}, & c &= e^{i\theta_{13}}, & d &= e^{i\theta_{03}}, \end{aligned} \quad (2.9)$$

Then

$$\tau^2 = abcd. \quad (2.10)$$

Exponentiating (2.8) gives

$$(p, a, b, q, c, d, \tau) \longmapsto \left(p, \frac{\tau}{a}, \frac{\tau}{b}, q, \frac{\tau}{c}, \frac{\tau}{d}, \tau\right). \quad (2.11)$$

This is the exact meaning of the phrase “the Regge transformation acts monomially.” A Laurent monomial in variables $z_1, \dots, z_m$ is a product $z_1^{n_1} \dots z_m^{n_m}$ with integral exponents. Every new coordinate in (2.11) is such a Laurent monomial. The additive formula $S - \theta$ has become the multiplicative formula τ/a . Thus no choice of logarithm occurs in the algebraic formula.

To make that observation scheme-theoretic, set

$$\mathcal{X} = \operatorname{Spec} \frac{\mathbb{Q}[p^{\pm 1}, a^{\pm 1}, b^{\pm 1}, q^{\pm 1}, c^{\pm 1}, d^{\pm 1}, \tau^{\pm 1}]}{(\tau^2 - abcd)}. \quad (2.12)$$

Here $\operatorname{Spec} R$ means the affine algebraic space whose regular functions form the ring R , and $\mathbf{G}_m = \operatorname{Spec} \mathbb{Q}[u, u^{-1}]$ is the affine line with the point $u = 0$ removed. Thus an F -point of $\mathcal{X}$, for a field $F \supset \mathbb{Q}$, is simply a tuple of nonzero elements of F satisfying $\tau^2 = abcd$. This notation packages polynomial identities and changes of coordinates; no extra geometric hypothesis is hidden in it. Eliminating $d = \tau^2/(abc)$ identifies $\mathcal{X}$ with the six-dimensional algebraic torus $\mathbf{G}_m^6$. Physical phase data form a real-analytic subset of $\mathcal{X}(\mathbb{C})$, characterized in part by $|p| = |a| = \cdots = |q| = 1$. We enlarge to all complex points because $\mathcal{X}$, unlike the physical locus, is an algebraic parameter space defined over $\mathbb{Q}$.

Lemma 2.4 (Algebraic Regge involution). *Formula (2.11) defines an involutive automorphism $\mathcal{R} : \mathcal{X} \rightarrow \mathcal{X}$.*

Proof. The product of the four transformed moving phases is

$$\frac{\tau^4}{abcd} = \tau^2,$$

so the defining relation of $\mathcal{X}$ is preserved. Applying the map twice sends, for example, a to $\tau/(\tau/a) = a$, and similarly for the other coordinates. The fixed coordinates remain fixed. $\square$

The phase $u = e^{i\theta}$ recovers the corresponding Gram entry without choosing θ :

$$-\cos \theta = -\frac{u + u^{-1}}{2}. \quad (2.13)$$

Accordingly put

$$\begin{aligned} x &= -\frac{p + p^{-1}}{2}, & A &= -\frac{a + a^{-1}}{2}, & B &= -\frac{b + b^{-1}}{2}, \\ y &= -\frac{q + q^{-1}}{2}, & C &= -\frac{c + c^{-1}}{2}, & D &= -\frac{d + d^{-1}}{2}. \end{aligned} \quad (2.14)$$

In the order fixed above, the universal angle Gram matrix is

$$G = \begin{pmatrix} 1 & x & A & D \\ x & 1 & B & C \\ A & B & 1 & y \\ D & C & y & 1 \end{pmatrix}. \quad (2.15)$$

At a physical point this specializes to (2.2).

2.3. The determinant orientation cover. On the open subset of $\mathcal{X}$ where $\det G \neq 0$, adjoin a variable w satisfying

$$w^2 = \det G. \quad (2.16)$$

Because the characteristic is zero and $w \neq 0$, this is a finite étale double cover. Its deck involution is $w \mapsto -w$. A finite étale map is the algebraic analogue of a finite covering without branch points. Here this is elementary: the defining polynomial is $w^2 - \det G$, and its derivative $2w$ is invertible on the chosen open, so its two roots never collide.

The terminology “orientation cover” refers to the projective quadric which will be attached to G later. A smooth quadric surface over an algebraically closed field has two families of maximal isotropic lines. For a four-dimensional quadratic form, choosing a square root of its determinant distinguishes these two families; changing the sign of the root interchanges them. Thus w , rather than the matrix G alone, retains the orientation needed by the quadric-scissors construction. At a physical spherical point we use the positive real root $w = \sqrt{\det G}$.

The determinant is Regge invariant. We now prove this in a form which also rationalizes the orientation cover.

Keep a, b, c, τ, q as independent invertible variables and eliminate

$$d = \frac{\tau^2}{abc}. \quad (2.17)$$

We temporarily forget the phase p , retaining only its cosine coordinate x . Define

$$s = \frac{q - q^{-1}}{2}, \quad y = -\frac{q + q^{-1}}{2}. \quad (2.18)$$

Then

$$1 - y^2 = -s^2. \quad (2.19)$$

The asymmetry between p and q is deliberate: we descend p to x , while the retained phase q supplies the square root s in (2.19).

Define

$$\mathcal{B} = 2\{AB + DC - y(AC + DB)\}, \quad (2.20)$$

$$\mathcal{C} = 1 - A^2 - B^2 - C^2 - D^2 - y^2 + 2y(AD + BC) + (AC - BD)^2. \quad (2.21)$$

Lemma 2.5 (Determinant quadratic and Regge invariance). *For the matrix (2.15),*

$$\det G = s^2 x^2 + \mathcal{B}x + \mathcal{C}. \quad (2.22)$$

Under the monomial Regge substitution

$$(a, b, c, d) \mapsto \left(\frac{\tau}{a}, \frac{\tau}{b}, \frac{\tau}{c}, \frac{\tau}{d} \right), \quad (2.23)$$

with p, q, τ fixed, one has

$$\mathcal{B}^R = \mathcal{B}, \quad \mathcal{C}^R = \mathcal{C}. \quad (2.24)$$

Consequently $\det G^R = \det G$, and the Regge involution has the lift

$$(G, w) \mapsto (G^R, w) \quad (2.25)$$

which does not change the chosen determinant root.

Proof. Expand the determinant of (2.15) as a polynomial in x . Its quadratic coefficient is $-(1-y^2) = s^2$, its linear coefficient is $2\{AB+DC-y(AC+DB)\}$, and its constant coefficient is the expression in (2.21). This proves (2.22).

For the invariance assertion, put

$$h(u) = \frac{u + u^{-1}}{2}$$

and introduce

$$\begin{aligned} I_1 &= AB + CD, & I_2 &= AC + BD, & I_3 &= AD + BC, \\ K &= A^2 + B^2 + C^2 + D^2 + 4ABCD. \end{aligned}$$

The elementary product-to-sum identity

$$2h(u)h(v) = h(uv) + h(u/v)$$

is an identity of Laurent polynomials. Under $a' = \tau/a, \dots, d' = \tau/d$, the relation $abcd = \tau^2$ gives

$$a'b' = cd, \quad c'd' = ab, \quad a'/b' = b/a, \quad c'/d' = d/c.$$

Since $h(u^{-1}) = h(u)$, the product-to-sum identity shows that $I_1^R = I_1$. Applying the same argument to the partitions $ac \mid bd$ and $ad \mid bc$ gives $I_2^R = I_2$ and $I_3^R = I_3$.

It remains to treat K . Set

$$P = h(ab), \quad Q = h(cd), \quad r = h(a/b), \quad s_0 = h(c/d).$$

Direct applications of the same product-to-sum identity give

$$\begin{aligned} h(a)^2 + h(b)^2 &= 1 + Pr, \\ h(c)^2 + h(d)^2 &= 1 + Qs_0, \\ 4h(a)h(b)h(c)h(d) &= (P+r)(Q+s_0). \end{aligned}$$

Because $A = -h(a), \dots, D = -h(d)$, it follows that

$$K = 2 + PQ + rs_0 + (P+Q)(r+s_0).$$

The Regge substitution swaps P, Q and fixes r, s_0 , so $K^R = K$. Finally,

$$\frac{\mathcal{B}}{2} = I_1 - yI_2, \quad \mathcal{C} = 1 - y^2 + 2yI_3 + I_2^2 - K,$$

where the second equality uses $(AC - BD)^2 = (AC + BD)^2 - 4ABCD$. Thus $\mathcal{B}^R = \mathcal{B}$ and $\mathcal{C}^R = \mathcal{C}$ inside the Laurent coordinate ring of $\mathcal{X}$, not merely on the physical locus. Since x, y, s are fixed, the determinant quadratics coincide. Keeping the same root w therefore gives the displayed lift. $\square$

2.4. Rational charts on the orientation cover. Let

$$S_0 = \mathbf{Q}[a^{\pm 1}, b^{\pm 1}, c^{\pm 1}, \tau^{\pm 1}, q^{\pm 1}],$$

and regard $d, A, B, C, D, y, s, \mathcal{B}, \mathcal{C}$ as the elements of S_0 defined above. The descended orientation space is the affine hypersurface

$$\mathcal{Y} = \text{Spec} \frac{S_0[x, w]}{(w^2 - s^2 x^2 - \mathcal{B}x - \mathcal{C})}. \quad (2.26)$$

The orientation cover of the phase torus maps to $\mathcal{Y}$ by $x = -(p + p^{-1})/2$. The point of passing to $\mathcal{Y}$ is that it is rational, although the original equation visibly contains a square root.

Proposition 2.6 (Two rational orientation charts). *Put*

$$r = w - sx, \quad r_+ = w + sx. \quad (2.27)$$

On the principal open set where $2sr - \mathcal{B} \neq 0$, one has

$$x = \frac{\mathcal{C} - r^2}{2sr - \mathcal{B}}, \quad w = r + sx. \quad (2.28)$$

On the principal open set where $-2sr_+ - \mathcal{B} \neq 0$, one has

$$x = \frac{\mathcal{C} - r_+^2}{-2sr_+ - \mathcal{B}}, \quad w = r_+ - sx. \quad (2.29)$$

Each formula identifies the indicated part of $\mathcal{Y}$ with an open subset of $\mathbf{G}_m^5 \times \mathbf{A}^1$. In particular, its function field is purely transcendental:

$$\mathbf{Q}(a, b, c, \tau, q, r) \quad \text{or} \quad \mathbf{Q}(a, b, c, \tau, q, r_+). \quad (2.30)$$

On the locus $sw \neq 0$, the two charts cover $\mathcal{Y}$.

Proof. Substitute $w = r + sx$ into (2.22):

$$r^2 + 2rsx + s^2 x^2 = s^2 x^2 + \mathcal{B}x + \mathcal{C}.$$

After cancellation of the quadratic terms,

$$(2sr - \mathcal{B})x = \mathcal{C} - r^2,$$

which is (2.28). Conversely, the two formulas in (2.28) satisfy both $r = w - sx$ and the defining equation of $\mathcal{Y}$; hence they give mutually inverse coordinate maps on the stated principal opens. The proof of (2.29) is identical after substituting $w = r_+ - sx$.

It remains to check coverage. Written in the original coordinates, the two denominators are

$$2s(w - sx) - \mathcal{B}, \quad -2s(w + sx) - \mathcal{B}.$$

If both vanished, subtracting the second equation from the first would give $4sw = 0$. This is impossible on the locus $sw \neq 0$. $\square$

The change of variable has a simple geometric-algebraic interpretation. The equation $w^2 = s^2x^2 + \mathcal{B}x + \mathcal{C}$ is a conic in the (x, w) -plane over the field $\mathbf{Q}(a, b, c, \tau, q)$. The coordinates $r = w - sx$ and $r_+ = w + sx$ subtract one or the other asymptotic linear part. This cancels the x^2 -term and leaves an equation linear in x . No continuity or density argument is involved.

By Lemma 2.5, the Regge involution fixes $x, w, q, s, \mathcal{B}, \mathcal{C}$. It therefore fixes r and r_+ and preserves both rational charts. The two oriented Gram matrices (G, w) and (G^R, w) are defined over the same chart.

2.5. The universal open set. The rational chart contains points which do not represent nondegenerate tetrahedra, and it also contains matrices with degenerate coordinate restrictions. These unwanted points lie on algebraic divisors and may be removed once and for all.

Fix one of the two charts in Proposition 2.6.

For a regular function f , the *principal open* $D(f)$ is obtained by removing its zero set, equivalently by adjoining f^{-1} to the coordinate ring. The *function field* of an irreducible affine variety is the field of fractions of its coordinate ring. Shrinking to a nonempty principal open changes the ring of regular functions but not this field of rational functions.

Definition 2.7 (Universal open). Let U be the Regge-stable principal open obtained by requiring:

- (1) the displayed chart denominator, s, w , and $1 - x^2$ are nonzero;
- (2) G and G^R are nonsingular; and
- (3) every proper principal restriction of G and G^R which occurs as a coordinate face is nonsingular.

Equivalently, take the product of these finitely many functions and their Regge transforms and invert that product. This makes the definition manifestly Regge stable. Subsequent constructions may shrink U further by finitely many Regge-stable nonvanishing conditions; such a shrinking does not change its function field.

Since a rational chart is an open subset of $\mathbf{G}_m^5 \times \mathbf{A}^1$, every nonempty such U is smooth and irreducible, and

$$\mathbf{Q}(U) = \mathbf{Q}(a, b, c, \tau, q, r) \quad \text{or} \quad \mathbf{Q}(a, b, c, \tau, q, r_+). \quad (2.31)$$

The word “universal” does not mean that every complex point of U is a convex spherical tetrahedron. It means that G, G^R , their common orientation root, and the Regge involution are all defined simultaneously over one algebraic base. The physical spherical pairs form a distinguished real locus inside its complex points.

Proposition 2.8 (Every physical pair lies on a rational chart). *Let $T, T^R \subset S^3$ be a nondegenerate spherical Regge pair. Choose the physical phases (2.9) and the positive root $w = \sqrt{\det G(T)}$. Then the resulting oriented point belongs to at least one of the two rational charts. After shrinking that chart*

as above, it is a complex point of a universal open U , and its Regge mate is the image of that point under $\mathcal{R}$.

Proof. Because $0 < \theta_{23} < \pi$,

$$s = \frac{e^{i\theta_{23}} - e^{-i\theta_{23}}}{2} = i \sin \theta_{23} \neq 0.$$

Nondegeneracy and positive definiteness give $w > 0$. Thus $sw \neq 0$, and Proposition 2.6 puts the point on at least one chart. All principal restrictions of a positive-definite matrix are positive definite; the same applies to the nondegenerate spherical mate. Hence none of the functions removed in the definition of U vanishes at the pair. The common-root lift (2.25) identifies the second oriented matrix with the Regge image of the first. $\square$

This is the step which removes any algebraicity restriction on an individual tetrahedron. Its angles may be transcendental: they merely specify a complex point of the algebraic family U . The later argument will prove one identity over U and then pull that identity back to this point.

2.6. Recovering the omitted fixed phase. The functions $a, b, c, q, d = \tau^2/(abc)$ are already defined on U . The only original angle phase which is not a function on U is p , because the rational chart remembers only

$$x = -\frac{p + p^{-1}}{2}.$$

Equivalently, p satisfies

$$p^2 + 2xp + 1 = 0. \tag{2.32}$$

Proposition 2.9 (The fixed-phase cover). *Define*

$$\pi : U_p = \operatorname{Spec} \frac{\mathcal{O}_U[p]}{(p^2 + 2xp + 1)} \longrightarrow U. \tag{2.33}$$

Then π is finite étale of degree two. Its nontrivial deck transformation is

$$p \longmapsto p^{-1} = -2x - p. \tag{2.34}$$

The cover is common to G and G^R , because the Regge transformation fixes the 01-angle and hence fixes x .

Proof. The polynomial in (2.32) is monic of degree two, so U_p is finite locally free of rank two over U . Its discriminant is

$$(2x)^2 - 4 = 4(x^2 - 1). \tag{2.35}$$

The function $1 - x^2$ was inverted in the definition of U ; hence the discriminant is a unit and the cover is étale. The product of the two roots is one, so the second root is p^{-1} , and their sum is $-2x$, giving (2.34). Finally, x is fixed by the Regge involution. $\square$

On a physical fibre the two points of U_p are $e^{i\theta_{01}}$ and $e^{-i\theta_{01}}$. The cover may split after base change, but it need not split over U itself. This two-sheeted fixed-phase datum is the sole residual effect of descending from the full phase torus to the rational orientation chart.

3. THE MOTIVIC DICTIONARY USED IN THE PROOF

This section is a dictionary, not a second proof. It explains the objects which occur in the motivic step and records exactly which features of the general theory are used. A reader may take the properties listed below as axioms for the argument. In particular, we do not require a conjectural abelian category of all mixed motives.

3.1. From a spherical simplex to an oriented quadric simplex. Let U be a smooth connected variety over $\mathbb{Q}$, and let E be a rank-four vector bundle on U . A nowhere-degenerate quadratic form on E cuts out a smooth relative quadric

$$Q \subset \mathbf{P}(E).$$

Let $M = (M_0, M_1, M_2, M_3)$ be four ordered relative hyperplanes in general position. We require the intersections of Q with all coordinate faces to have the expected dimension and to be regular. The pair (Q, M) is the algebro-geometric version of a tetrahedron: the quadric is the complexified null cone which encodes the spherical metric (classically called the *absolute*), and the M_i are its four faces.

In the family constructed in Section 2 this data is concrete:

$$E = \mathcal{O}_U^4, \quad M_i = \{z_i = 0\}.$$

The spherical simplex gives the quadratic form with matrix G^{-1} , hence $Q_{G^{-1}}$; projective duality replaces it by Q_G , on which the coordinate-face calculation is easier. The function w , satisfying $w^2 = \det G$, chooses one of the two rulings and therefore orients both descriptions.

For a quadric surface, the maximal isotropic subspaces are lines. Over an algebraic closure they occur in two rulings. An *orientation* of Q is a choice of one of these two components, made compatibly in the family. For a matrix presentation

$$Q_G : \quad z^t G z = 0,$$

the two choices are distinguished by a square root w of $\det G$. Changing w interchanges the rulings and reverses the oriented quadric-simplex class. Thus the natural parameter space for oriented quadrics is the determinant double cover $w^2 = \det G$, rather than the space of matrices G alone.

The notation

$$[(Q, \alpha); M_0, M_1, M_2, M_3]$$

means the resulting oriented quadric simplex, where α is the chosen ruling. Goncharov's group $S_4(F)$, for a field F , is generated by such objects over F , modulo projective invariance, alternating face relations, and the standard

additivity relations. We will not need a presentation of all these relations; we use only its face coproduct and the canonical maps listed in Section 3.9.

There are two equivalent matrix descriptions which occur in the proof. Start with

$$Q_{G^{-1}} : \quad x^t G^{-1} x = 0 \subset \mathbf{P}(E)$$

and the coordinate simplex. Goncharov's projective-duality operation replaces the face opposite the vertex e_j by its polar hyperplane

$$e_j^t G^{-1} x = 0.$$

Now make the projective change of coordinates $x = Gy$. The quadric equation becomes $y^t G y = 0$, and the four polar hyperplanes become $y_j = 0$. Thus the dual class is represented by Q_G with the coordinate faces. This operation is an involutive linear automorphism of quadric-scissors classes, so a class is zero if and only if its dual is zero. This is why the face calculation may be performed on Q_G , even though the quadric first attached to the Gram matrix is $Q_{G^{-1}}$. The determinant root transports the orientation through the operation.

3.2. What a relative motive means here. A cohomology theory replaces a variety by groups. A motive retains the common algebraic structure which maps to all the usual realizations—for example Betti, de Rham, or Hodge cohomology. In this proof it is important that a motive is an *object*, not merely a numerical invariant: it can be put into exact triangles, dualized, filtered, and pulled back to a point.

We work in the rational constructible motivic category

$$\mathrm{DM}(U)_{\mathbb{Q}}.$$

For the present argument this may be understood as a symmetric monoidal, idempotent-complete triangulated category with the following operations. Our cohomological shift convention is $H^i(A[m]) = H^{i+m}(A)$ in every realization. A distinguished triangle $A \rightarrow B \rightarrow C \rightarrow A[1]$ is the derived replacement for a short exact sequence: every cohomology realization turns it into a long exact sequence. Thus the triangles below are not extra geometric objects; they are the bookkeeping device which remembers kernels, quotients, and connecting maps simultaneously.

- (i) It has a unit object $\mathbb{Q}_U$, shifts $A[m]$, tensor products, and internal Homs $\underline{\mathrm{Hom}}(A, B)$.
- (ii) A smooth U -scheme $f : X \rightarrow U$ has a geometric motive $M_U(X)$. Disjoint unions become direct sums and $M_U(U) = \mathbb{Q}_U$.
- (iii) If $Z \hookrightarrow X$ is a regular closed immersion with open complement V , localization gives a distinguished triangle. With purity, for codimension c it can be written in Gysin form

$$M_U(V) \longrightarrow M_U(X) \longrightarrow M_U(Z)(c)[2c] \longrightarrow M_U(V)[1].$$

Repeated localization is the motivic version of cutting a space into strata.

- (iv) Constructible objects in the calculation are dualizable. We write

$$D(A) = \underline{\mathrm{Hom}}(A, \mathbb{Q}_U).$$

Poincaré duality relates this internal dual to compactly supported cohomology in every realization.

- (v) For a geometric point $s : \mathrm{Spec} k \rightarrow U$, exact symmetric monoidal pullback

$$s^* : \mathrm{DM}(U)_{\mathbb{Q}} \longrightarrow \mathrm{DM}(k)_{\mathbb{Q}}$$

commutes with every construction used below: geometric motives, localization maps, tensor products, twists, idempotent summands, and internal duals.

- (vi) A finite étale map has pullback, pushforward, a unit, and a trace. For a degree- d cover, trace after the unit is multiplication by d . This is the only finite-étale descent fact used below.

These are the only formal properties of the ambient motivic category that enter the proof.

For later use, *motivic cohomology* is simply the following Hom group in this category:

$$H^m(U, \mathbb{Q}(n)) = \mathrm{Hom}_{\mathrm{DM}(U)_{\mathbb{Q}}}(\mathbb{Q}_U(0), \mathbb{Q}_U(n)[m]). \quad (3.1)$$

Thus every motivic-cohomology class used in the proof is an actual morphism between two explicitly named objects.

3.3. Tate objects and Artin objects. The *Tate object* $\mathbb{Q}_U(1)$ is the basic one-dimensional object detected by the multiplicative group. Our convention is

$$\mathbb{Q}(1) = \mathbf{G}_m[-1]$$

where on the right $\mathbf{G}_m$ denotes the motivic sheaf represented by the multiplicative group; this is an equality of motivic objects, not of schemes. Concretely, on every smooth connected scheme W used here,

$$H^1(W, \mathbb{Q}(1)) \cong \mathcal{O}(W)^* \otimes_{\mathbf{Z}} \mathbb{Q}. \quad (3.2)$$

This is the bridge between the angle and length cross-ratios, which are invertible functions, and the middle Tate degree of the motive. For any object A , its n -fold Tate twist is

$$A(n) = A \otimes \mathbb{Q}_U(n), \quad \mathbb{Q}_U(n)^{\vee} = \mathbb{Q}_U(-n).$$

The integers 0, 1, 2 used below are *Tate degrees*. They are the three levels relevant to a three-dimensional quadric simplex; they should not be confused with the signs or factor-of-two conventions used for Deligne's numerical weights.

An *Artin motive* is the motivic avatar of a finite étale cover. The only nonsplit one needed here comes from a quadratic cover

$$\pi : U' \longrightarrow U$$

with deck involution σ . Rationally,

$$M_U(U') = \pi_* \mathbb{Q}_{U'} \cong \mathbb{Q}_U \oplus \chi, \quad \chi = \text{im } \frac{1 - \sigma}{2}. \quad (3.3)$$

The first summand is invariant and the second is the sign motive. More intrinsically, if $D \rightarrow U$ is any degree-two finite étale scheme, its reduced motive is the idempotent summand

$$\widetilde{M}(D) = \text{im} \left(1 - \frac{1}{2} \eta \text{Tr} : M_U(D) \longrightarrow M_U(D) \right), \quad (3.4)$$

where η is the unit and Tr is the trace. For a connected quadratic cover this is χ . For a split pair of sections it is a copy of $\mathbb{Q}_U$. The rational Artin category is semisimple: for a Galois cover this is simply Maschke's theorem for rational representations of its finite deck group.

3.4. The alternating face complex.

Definition 3.1 (Normalized relative quadric motive). For $I \subset \{0, 1, 2, 3\}$, put

$$M_I = \bigcap_{i \in I} M_i, \quad Q_I = Q \cap M_I.$$

The order of the hyperplanes supplies the usual alternating incidence signs. Place the $|I| = 3$ term in cohomological degree 0, and the following three terms in degrees 1, 2, 3, respectively. The geometric face complex is the total object

$$\begin{aligned} \bigoplus_{|I|=3} M_U(M_I \setminus Q_I) &\longrightarrow \bigoplus_{|I|=2} M_U(M_I \setminus Q_I) \\ &\longrightarrow \bigoplus_{|I|=1} M_U(M_I \setminus Q_I) \longrightarrow M_U(\mathbf{P}(E) \setminus Q). \end{aligned} \quad (3.5)$$

The phrase *total object* means the finite totalization of this four-term complex. We use the standard stable (equivalently, dg) enhancement of $\text{DM}(U)_{\mathbb{Q}}$, in which a bounded diagram has a functorial totalization. Its image in the triangulated homotopy category may be computed by iterated mapping cones. We denote it by $\widetilde{m}_U(Q, M)$. Its raw cohomological dual and its normalized version are

$$\begin{aligned} h_U(Q, M) &= D(\widetilde{m}_U(Q, M)), \\ \mathfrak{m}_U(Q, M) &= h_U(Q, M)(2) \\ &= \underline{\text{Hom}}(\widetilde{m}_U(Q, M), \mathbb{Q}_U)(2). \end{aligned} \quad (3.6)$$

The face complex is the motivic analogue of the relative cellular complex of the complement of Q , relative to its four ordered faces. Its alternating signs record how a face occurs in the boundary of the next-dimensional face. This description is useful because localization can be applied to each displayed stratum separately.

The twist (2) in Definition 3.1 is essential and there is no additional cohomological shift. Before twisting, the two endpoints have Tate degrees 0

and -2 ; after twisting they become 2 and 0 . This is precisely the normalization in which a primitive endpoint is an extension of $\mathbb{Q}_U(0)$ by $\mathbb{Q}_U(2)$.

3.5. Why only three weights occur. Assume $Q \subset \mathbf{P}_U^3$ is oriented and split, every plane section $Q \cap M_i$ is a split regular conic, and every line section

$$D_I = Q \cap M_I \subset M_I \cong \mathbf{P}_U^1, \quad |I| = 2,$$

is finite étale of degree two. Localization, purity, and the projective bundle formula give

$$M_U(\mathbf{P}_U^3 \setminus Q) \cong \mathbb{Q}_U \oplus \mathbb{Q}_U(2)[3], \quad (3.7)$$

$$M_U(\mathbf{P}_U^2 \setminus C) \cong \mathbb{Q}_U \quad (C \text{ a split regular conic}), \quad (3.8)$$

$$M_U(\mathbf{P}_U^1 \setminus D_I) \cong \mathbb{Q}_U \oplus \widetilde{M}(D_I)(1)[1], \quad (3.9)$$

$$M_U(\mathbf{P}_U^0) \cong \mathbb{Q}_U. \quad (3.10)$$

The shifts $[3]$ and $[1]$ cancel the positions of the corresponding terms in the face complex. After internal duality and the twist (2) , the normalized relative motive therefore has a three-step filtration with graded pieces

$$\mathrm{gr}^0 \mathbf{m}_U \cong \mathbb{Q}_U(0), \quad \mathrm{gr}^1 \mathbf{m}_U \cong \bigoplus_{|I|=2} \widetilde{M}(D_I)(1), \quad \mathrm{gr}^2 \mathbf{m}_U \cong \mathbb{Q}_U(2). \quad (3.11)$$

We use the decreasing convention $F^0 \supset F^1 \supset F^2 \supset F^3 = 0$, with $\mathrm{gr}_F^i = F^i/F^{i+1}$. Thus $\mathbb{Q}_U(2)$ is the bottom subobject F^2 , while $\mathbb{Q}_U(0)$ is the top quotient F^0/F^1 . This convention fixes the direction of the endpoint extension used below. This formula is the reason the argument is called three-weight. All other summands in the dual cellular row cancel, and the three displayed kinds of objects are the complete list of its graded pieces. If five binary sections split and one is the quadratic cover $U' \rightarrow U$, then the middle term is a direct sum of copies of $\mathbb{Q}_U(1)$ and one possible copy of $\chi(1)$.

3.6. The restricted heart and extensions. One often packages mixed motives into an abelian category, the *heart* of a motivic t -structure. A t -structure is a rule separating a triangulated category into nonpositive and nonnegative parts; their intersection is an abelian category, and triangles among its objects are short exact sequences. The existence of such a heart for every motive over every base is not assumed here. We use only the finite window generated by

$$\mathcal{P}_0 = \langle \mathbb{Q}_U(0) \rangle, \quad \mathcal{P}_1 = \langle \mathbb{Q}_U(1), \chi(1) \rangle, \quad \mathcal{P}_2 = \langle \mathbb{Q}_U(2) \rangle. \quad (3.12)$$

Each $\mathcal{P}_i$ is semisimple. If the negative Hom groups between these three pure levels vanish, the finite semisimple gluing theorem equips their thick triangulated closure with a bounded t -structure. Its heart, denoted

$$\mathcal{M}_U^{[0,2]},$$

is the extension closure of the three $\mathcal{P}_i$. Here “thick” means closed under shifts, cones, finite sums, and direct summands; “extension closure” means objects

built from the displayed pure objects by finitely many short exact sequences; and a negative Hom means $\text{Hom}(P_i, P_j[m])$ with $m < 0$. Concretely, its objects are assembled from the three pure levels by finitely many short exact sequences. A distinguished triangle whose three terms lie in the heart is the corresponding short exact sequence. To obtain the functorial three-step filtration one also uses the vanishing of wrong-way degree-zero maps and degree-one extensions. Any extension written in the wrong weight order then splits and may be reordered; induction on extension length gives a unique maximal subobject of weights at least i . These subobjects are F^i , and their uniqueness makes every morphism strict, meaning $f(F^i M) = \text{im}(f) \cap F^i N$ for every morphism $f : M \rightarrow N$. Section 4 verifies these additional vanishings in the present motivic window. Inside the restricted heart, the heart-to-derived comparison gives

$$\text{Ext}_{\mathcal{M}_U^{[0,2]}}^1(\mathbb{Q}_U(0), \mathbb{Q}_U(2)) \cong H^1(U, \mathbb{Q}(2)). \quad (3.13)$$

An element of this group is represented by an exact sequence

$$0 \longrightarrow \mathbb{Q}_U(2) \longrightarrow E \longrightarrow \mathbb{Q}_U(0) \longrightarrow 0,$$

or equivalently by the endpoint morphism $\mathbb{Q}_U(0) \rightarrow \mathbb{Q}_U(2)[1]$ in its distinguished triangle.

3.7. Frames, coproducts, and primitive classes.

Definition 3.2 (Framed class). A $(0, 2)$ -framing on an object of $\mathcal{M}_U^{[0,2]}$ specifies its two endpoints: a chosen identification of its degree-zero graded piece with $\mathbb{Q}_U(0)$, and a chosen identification of its degree-two graded piece with $\mathbb{Q}_U(2)$. Equivalently, after applying the graded fibre functor defined below, one gives the corresponding nonzero top vector and bottom covector. Framed equivalence remembers these maps while allowing changes to the unframed middle of the object.

A mixed object alone forgets which geometric fundamental classes are being compared. For $\mathfrak{m}_U(Q, M)$, the chosen ruling of the oriented quadric supplies the top $\mathbb{Q}_U(0)$ -vector, and the ordering and orientation of the four coordinate faces supplies the bottom $\mathbb{Q}_U(2)$ -covector. Thus an oriented quadric simplex determines a framed class, not just an unmarked motive.

Definition 3.3 (Coefficient spaces). Let X be simple in $\mathcal{P}_i$, let Y be simple in $\mathcal{P}_j$, and assume $i < j$. An (X, Y) -framed coefficient is a triple

$$(M, v, f), \quad v : X \longrightarrow \text{gr}_F^i M, \quad f : \text{gr}_F^j M \longrightarrow Y.$$

Thus v is the top vector and f the bottom covector. Two triples are identified when a morphism between their middle objects carries both frames to one another; one takes the equivalence relation generated by such morphisms, makes the construction linear in the frames, and uses direct sum for addition. The resulting rational vector space is denoted $A(X, Y)$. We write $\text{Irr}(\mathcal{P}_i)$ for representatives of the simple objects of $\mathcal{P}_i$.

The following length-three statement is the precise result behind the phrase *primitive equals an extension*.

Lemma 3.4 (Three-level framed-object lemma). *Let $\mathcal{C}$ be an essentially small rational finite-length abelian category with finite-dimensional Hom spaces and a functorial strict decreasing filtration*

$$M = F^0 M \supset F^1 M \supset F^2 M \supset F^3 M = 0, \quad \mathrm{gr}_F^i M \in \mathcal{P}_i,$$

where the $\mathcal{P}_i$ are semisimple. Suppose each $\mathcal{P}_i$ has an exact faithful functor to finite-dimensional rational vector spaces. For simple $X \in \mathcal{P}_0$ and $Y \in \mathcal{P}_2$,

$$\ker \left(\bar{\Delta}_{X,Y} : A(X, Y) \longrightarrow \bigoplus_{S \in \mathrm{Irr}(\mathcal{P}_1)} A(X, S) \otimes_{\mathrm{End}(S)} A(S, Y) \right) \cong \mathrm{Ext}_{\mathcal{C}}^1(X, Y). \quad (3.14)$$

An extension $0 \rightarrow Y \rightarrow E \rightarrow X \rightarrow 0$ corresponds to E framed by the identity maps on its two graded pieces.

Proof. Strictness makes the associated-graded functor exact. It is faithful as well: if every graded map of a morphism is zero, strictness says that its image has all graded pieces zero, hence the image itself is zero. Composing with the chosen fibre functors gives an exact faithful functor

$$\omega = \bigoplus_{i=0}^2 \omega_i \mathrm{gr}_F^i : \mathcal{C} \longrightarrow \mathrm{Vect}_{\mathbb{Q}}^{\mathrm{fd}}.$$

We recall the coefficient-coalgebra construction. Start with

$$\bigoplus_{M \in \mathcal{C}} \omega(M)^{\vee} \otimes \omega(M)$$

and impose, for every $g : M \rightarrow N$, the matrix-coefficient relation

$$[\omega(g)^{\vee} \lambda \otimes x]_M = [\lambda \otimes \omega(g)x]_N.$$

The quotient is a coalgebra: if (e_a) and $(e_a^{\vee})$ are dual bases, its coproduct is

$$[\lambda \otimes x]_M \longmapsto \sum_a [\lambda \otimes e_a]_M \otimes [e_a^{\vee} \otimes x]_M.$$

This quotient is also called the *coend* of ω . A coalgebra is a vector space with a coassociative map to its tensor square; a comodule is a vector space carrying a compatible coaction. The coefficient-coalgebra reconstruction theorem identifies $\mathcal{C}$ with its category of finite-dimensional comodules; it requires an exact faithful fibre functor, but no tensor product on $\mathcal{C}$. The spaces in Definition 3.3 are its source–target matrix-coefficient spaces: $A(X, Y)$ is the block cut out by the idempotents belonging to the pure simples X and Y .

Because the pure categories are semisimple, their coefficient coalgebra is cosemisimple, meaning that its finite-dimensional comodules are semisimple. The normalized cobar resolution is the standard complex obtained by repeatedly applying the coproduct after deleting the two pure endpoint terms; it computes Yoneda extensions. Thus the *reduced* coproduct below retains

exactly the factorizations through the middle level. In source X , target Y , and total level two, its first two positive-degree terms are

$$A(X, Y) \xrightarrow{\bar{\Delta}_{X,Y}} \bigoplus_{S \in \text{Irr}(\mathcal{P}_1)} A(X, S) \otimes_{\text{End}(S)} A(S, Y).$$

There is no incoming degree-zero term because $\text{Hom}_{\oplus \mathcal{P}_i}(X, Y) = 0$ for distinct pure levels. To see the displayed differential directly, choose on every S -isotypic part of $\text{gr}_F^1 M$ dual maps

$$u_a : S \longrightarrow \text{gr}_F^1 M, \quad u^a : \text{gr}_F^1 M \longrightarrow S.$$

Then

$$\bar{\Delta}_{X,Y}[M, v, f] = \sum_{S,a} [M/F^2 M, v, u^a] \otimes_{\text{End}(S)} [F^1 M, u_a, f]. \quad (3.15)$$

Thus degree-one cobar cohomology is the kernel in (3.14); reconstruction identifies it with $\text{Ext}_{\mathcal{C}}^1(X, Y)$. $\square$

For the category used here, take $X = \mathbb{Q}_U(0)$ and $Y = \mathbb{Q}_U(2)$. The pure fibre functors are the usual forgetful functors on rational vector spaces and on the rational Artin representations generated by χ . Once the vanishings giving the strict filtration have been verified, Lemma 3.4 yields

$$\ker \bar{\Delta}_{0,2} \cong \text{Ext}_{\mathcal{M}_U^{[0,2]}}^1(\mathbb{Q}_U(0), \mathbb{Q}_U(2)) \cong H^1(U, \mathbb{Q}(2)). \quad (3.16)$$

A framed class in this kernel is called *primitive*. In plain language, once every factorization through the middle level has cancelled, the only remaining obstruction is a direct extension of the top quotient by the bottom subobject.

For a quadric tetrahedron, the six line sections D_I give the six summands in (3.15). After a splitting cover their two half-frames are weight-one cross-ratios: the angle coordinate and the complementary edge coordinate. Section 4 computes their sum.

For a field F , weight-two motivic cohomology has the K -theoretic description

$$H^1(F, \mathbb{Q}(2)) \cong \text{gr}_2^\gamma K_3(F) \otimes \mathbb{Q}. \quad (3.17)$$

Here $K_3(F)$ is Quillen's third algebraic K -group and gr_2^γ is the second associated-graded piece of its standard gamma filtration. No other part of algebraic K -theory is used in the proof. This explains why K_3 appears at the end of a scissors-congruence argument: it is the group classifying the primitive endpoint extensions in Tate degree two.

3.8. Specialization is pullback. Suppose the family (Q, M, α) is defined over U , and let $s : \text{Spec } \mathbb{C} \rightarrow U$ be a complex point. Its specialization is not an appeal to analytic continuity. It is the exact functor s^* . Because every face term is constructible and dualizable, pullback gives

$$s^* \mathfrak{m}_U(Q, M) \cong \mathfrak{m}_{\mathbb{C}}(Q_s, M_s),$$

and preserves the two frames, the three-step filtration, its coproduct, and the endpoint triangle. In particular, a zero relative framed class pulls back to a zero framed class at every complex point.

If a primitive class is represented by

$$\mathbb{Q}_U(2) \longrightarrow E \longrightarrow \mathbb{Q}_U(0) \xrightarrow{\kappa_U} \mathbb{Q}_U(2)[1],$$

then its fibre is represented by the literally pulled-back triangle and endpoint $s^*\kappa_U$. This observation does not require a motivic t -structure over $\mathbb{C}$, nor does it require pullback to be t -exact: the endpoint morphism already exists in the triangulated category before pullback.

Goncharov's construction assigns to the specialized oriented quadric simplex the same normalized framed face motive. Hence its normalized-cobar endpoint is exactly the canonical class in

$$H^1(\mathbb{C}, \mathbb{Q}(2)) \cong \mathrm{gr}_2^\gamma K_3(\mathbb{C}) \otimes \mathbb{Q},$$

not merely a period or regulator of that class.

3.9. Standard inputs and the boundary of the argument. For clarity, the proof treats the following results as standard external inputs.

- (1) The rational constructible motivic formalism over the smooth bases in question: localization, purity, the projective bundle formula, internal duality, finite-étale trace, and exact symmetric monoidal pullback. We also use cancellation and the higher-Chow description of motivic cohomology, in particular the vanishing in negative Tate degree and $H^1(W, \mathbb{Q}(0)) = 0$ for the smooth schemes W occurring here.
- (2) Goncharov's finite semisimple gluing construction [Gon99, Chapter 5, Section 1], applied only to the three pure categories in Equation (3.12).
- (3) Goncharov's relative quadric motive [Gon99, Chapter 5, Section 2], together with the weight-coefficient and coproduct computations in [Gon99, Theorem 4.8 and Proposition 4.9]. The twist in Equation (3.6) agrees with the normalization in [Bro13, Definition 5.8].
- (4) The coefficient-coalgebra reconstruction theorem and its normalized cobar resolution [BGSV90, Gon99]. Their complete length-three specialization is proved in Lemma 3.4.
- (5) Garkusha's weight-two birational comparison and invariance under a purely transcendental extension [Gar23, Corollaries 3.10, 3.11, 3.13 and Theorem 3.14]. For the rational open base used later, these give, in degrees $m \leq 1$,

$$H^m(U, \mathbb{Q}(2)) \cong H^m(\mathbb{Q}(t_1, \dots, t_6), \mathbb{Q}(2)) \cong H^m(\mathbb{Q}, \mathbb{Q}(2)).$$

- (6) Beilinson–Soulé vanishing in the required nonpositive degrees over the number field $\mathbb{Q}$, and Borel's rank theorem [Bor77], which gives

$$H^1(\mathbb{Q}, \mathbb{Q}(2)) = \mathrm{gr}_2^\gamma K_3(\mathbb{Q}) \otimes \mathbb{Q} = 0.$$

Theorem 3.5 (Weight-two comparison). *The canonical map defined in Goncharov's equations (4.6)–(4.7) [Gon99] is valid and injective in weight two*

by his Theorem 4.13. His Theorem 4.14 supplies the compatible map from spherical scissors classes to quadric simplices; the injectivity used to return from a vanishing quadric class to a vanishing reduced spherical scissors class is stated immediately after the proof of that theorem.

The argument does *not* assume Goncharov's Conjecture 4.15, which asks for a stronger generator-level chain map from the full quadric complex to a Bloch complex. Nor does it infer an algebraic identity from equality of regulators. The Regge-specific work that follows has three tasks only: construct the common rational base, compute the middle coproduct and prove that it vanishes there, and identify the resulting fibre with the canonical class to which Theorem 3.5 applies.

4. THE UNIVERSAL REGGE DEFECT AND WEIGHT-TWO RIGIDITY

We now apply the language of Section 3 to the universal pair over the rational open U constructed in Section 2. The argument has two independent parts. First we calculate the weight pieces of the two relative quadric motives. Then we calculate their middle coproduct and use a vanishing theorem to show that no endpoint extension remains.

Fix the physical point chosen in Proposition 2.8. From now on we replace U , if necessary, by a smaller Regge-stable principal open which still contains that point. We require on this open that every cross-ratio used below is a regular unit and that the finitely many root covers used to choose isotropic directions and half-phases are finite étale. At the physical point the relevant phases are nonzero and the face sections are smooth, so none of these additional conditions removes the point. This shrinking changes neither the function field in (2.31) nor the birational motivic cohomology calculation.

4.1. Why the dual quadrics and all their faces split. Let G and G^R be the two universal angle Gram matrices over U , with their common determinant root w . The quadric simplex naturally attached to G uses the quadric $Q_{G^{-1}}$. Goncharov's projective duality, which is the antipode in the quadric-scissors Hopf algebra, replaces it by the coordinate simplex

$$Q_G : z^t G z = 0 \subset \mathbf{P}_U^3 \quad (4.1)$$

with the same four ordered coordinate hyperplanes. It is enough to kill the dual defect: the antipode is an involution. We use the dual because its coordinate sections can be read directly from the entries of G .

A quadratic form over a scheme is called *split* if it is a direct sum of hyperbolic planes, locally represented by matrices $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. A split quadric surface is fibrewise $\mathbf{P}^1 \times \mathbf{P}^1$, and its two projections are its two rulings by lines.

Lemma 4.1 (Splitting of the universal coordinate strata). *The quadrics Q_G and Q_{G^R} are split over U . Every coordinate plane section is a split conic. Of the six coordinate line sections, five are split degree-two schemes; the remaining section is the common finite étale cover $U_p \rightarrow U$ from (2.33).*

Proof. For a unit u , put $g = -(u + u^{-1})/2$. The vector $(1, u)$ is isotropic for the binary form

$$\begin{pmatrix} 1 & g \\ g & 1 \end{pmatrix},$$

because $1 + 2gu + u^2 = 0$. In particular, the binary block of G controlled by the phase q is a hyperbolic plane.

Orthogonally eliminate this nondegenerate binary block in the rank-four quadratic bundle. On the remaining two coordinate directions the restricted form is its binary Schur complement C . This complement is oriented by the ordered trivial bundle and the ordered isotropic basis of the q -plane. In the notation of Section 2,

$$\det G = w^2, \quad \det \begin{pmatrix} 1 & y \\ y & 1 \end{pmatrix} = 1 - y^2 = -s^2.$$

The determinant formula for a Schur complement therefore gives

$$\det C = -\left(\frac{w}{s}\right)^2.$$

Write $t = w/s$, a unit on U . In an oriented local basis of the complement put

$$J_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad J = t^{-1} J_0 C.$$

For every symmetric binary matrix C , $(J_0 C)^2 = -\det(C)I$. Hence $J^2 = I$, and a direct multiplication gives $J^t C + C J = 0$. Also $\text{tr } J = 0$, so the two idempotents $(1 \pm J)/2$ have rank one and their images are line bundles. If $Jv = \pm v$, the skew-adjoint identity gives $C(v, v) = -C(Jv, Jv) = -C(v, v)$, so each eigenline is isotropic. Nondegeneracy of C then pairs the two eigenlines perfectly. Thus the complement, and hence the rank-four form, is hyperbolic. This proves that Q_G is split. The mate has the same q, s, w , so the identical argument applies to Q_{GR} .

Every coordinate rank-three restriction contains a binary block controlled by at least one of a, b, c, d, q . It therefore has an isotropic line. A smooth conic with a rational point is split, proving the plane-section claim. A coordinate rank-two restriction has two isotropic points. The phases a, b, c, d, q display those points for five restrictions. The last restriction is defined by $p + p^{-1} = -2x$, and its two isotropic points are defined precisely over U_p . Regge fixes p, q and replaces the other four phases by $\tau/a, \tau/b, \tau/c, \tau/d$, so the same description holds for the mate. $\square$

4.2. The three weight pieces. For a coordinate line $M_I \cong \mathbf{P}_U^1$, let $D_I = Q_G \cap M_I$. It is a finite étale scheme of degree two. Its motive has a constant part and a reduced part. With η denoting the unit and Tr the trace, define the latter as the idempotent image

$$\widetilde{M}(D_I) = \text{im} \left(1 - \frac{1}{2} \eta \text{Tr} : M(D_I) \longrightarrow M(D_I) \right). \quad (4.2)$$

If $D_I = U \amalg U$, this is a copy of the constant motive $\mathbb{Q}_U$, carrying the difference of the two points. For the nonsplit p -section it is the sign Artin motive χ of the double cover U_p/U .

Proposition 4.2 (Weight calculation for the quadric face motive). *For $Q = Q_G$ or Q_{G^R} , the normalized relative motive $\mathbf{m}_U(Q, M)$ of Definition 3.1 has a three-step Postnikov tower with pure cones*

$$\begin{aligned} \mathrm{gr}_F^0 \mathbf{m}_U(Q, M) &= \mathbb{Q}_U(0), \\ \mathrm{gr}_F^1 \mathbf{m}_U(Q, M) &= \bigoplus_{|I|=2} \widetilde{M}(D_I)(1), \\ \mathrm{gr}_F^2 \mathbf{m}_U(Q, M) &= \mathbb{Q}_U(2). \end{aligned} \tag{4.3}$$

After the gluing in Lemma 4.3, the motive lies in the resulting heart and this tower is its strict three-step filtration. Here $F^0 \supset F^1 \supset F^2 \supset F^3 = 0$ is the decreasing Tate-degree filtration: $\mathbb{Q}_U(2) = F^2$ is the bottom subobject and $\mathbb{Q}_U(0) = F^0/F^1$ is the top quotient. Thus the middle piece is a sum of five copies of $\mathbb{Q}_U(1)$ and one copy of $\chi(1)$. The ruling selected by w gives the $\mathbb{Q}_U(0)$ -frame, and the orientation of the four ordered faces gives the $\mathbb{Q}_U(2)$ -frame.

Proof. We give the calculation because it is the only place where the geometric face complex is converted into weights. Localization and the projective-bundle formula give

$$M_U(\mathbf{P}_U^3 \setminus Q) \simeq \mathbb{Q}_U \oplus \mathbb{Q}_U(2)[3], \tag{4.4}$$

$$M_U(\mathbf{P}_U^2 \setminus C) \simeq \mathbb{Q}_U \quad (C \text{ a split smooth conic}), \tag{4.5}$$

$$M_U(\mathbf{P}_U^1 \setminus D_I) \simeq \mathbb{Q}_U \oplus \widetilde{M}(D_I)(1)[1], \tag{4.6}$$

$$M_U(\mathbf{P}_U^0) \simeq \mathbb{Q}_U. \tag{4.7}$$

Here is the relative calculation behind these formulas. A split quadric surface has the projective-bundle motive

$$M_U(Q) = \mathbb{Q}_U \oplus \mathbb{Q}_U(1)[2]^{\oplus 2} \oplus \mathbb{Q}_U(2)[4].$$

In the Gysin map to $M_U(\mathbf{P}_U^3)$, the two middle ruling classes map by their sum to the unique $\mathbb{Q}_U(2)[4]$ -class; their difference is the primitive class. The bottom and top maps are multiplication by the nonzero geometric degrees and split over $\mathbb{Q}$. The localization triangle therefore leaves exactly $\mathbb{Q}_U$ and the primitive $\mathbb{Q}_U(2)[3]$. The same argument for a split conic cancels its two nonconstant projective-bundle terms. For $D_I \subset \mathbf{P}_U^1$, the trace summand of $M_U(D_I)$ maps by degree 2, while the reduced idempotent summand maps to zero; shifting the latter across the localization triangle gives $\widetilde{M}(D_I)(1)[1]$. Thus the displayed decompositions are relative over U , not merely fibrewise identities.

For example, the localization triangle for the two-point divisor $D_I \subset \mathbf{P}_U^1$ splits its motive into the trace part and the reduced summand (4.2). In the

conic and binary localization triangles, the map on the constant degree-two part is multiplication by the geometric degree 2, and is therefore invertible over $\mathbb{Q}$. The rank-one primitive summand in (4.4) is the difference of the two ruling classes; the orientation root w selects its generator, hence its sign and frame.

Apply the dual-and-twist operation

$$D_2(-) = \underline{\mathrm{Hom}}(-, \mathbb{Q}_U)(2)$$

appearing in Definition 3.1. Trace self-duality of a degree-two étale scheme gives

$$\begin{aligned} D_2(M_U(\mathbf{P}_U^3 \setminus Q)) &\simeq \mathbb{Q}_U(2) \oplus \mathbb{Q}_U(0)[-3], \\ D_2(M_U(\mathbf{P}_U^2 \setminus C)) &\simeq \mathbb{Q}_U(2), \\ D_2(M_U(\mathbf{P}_U^1 \setminus D_I)) &\simeq \mathbb{Q}_U(2) \oplus \widetilde{M}(D_I)(1)[-1], \\ D_2(M_U(\mathbf{P}_U^0)) &\simeq \mathbb{Q}_U(2). \end{aligned}$$

The cohomological bookkeeping is short. The exceptional ambient summand occurs in face-complex degree three, so totalization changes $\mathbb{Q}_U(0)[-3]$ to $\mathbb{Q}_U(0)$. Each exceptional line summand occurs in degree one, changing $\widetilde{M}(D_I)(1)[-1]$ to $\widetilde{M}(D_I)(1)$. The remaining constant summands are handled by their cellular differential, as follows. The constant $\mathbb{Q}_U(2)$ -summands form the dual unaugmented cellular complex of the ordered tetrahedron,

$$\mathbb{Q}_U(2) \longrightarrow \mathbb{Q}_U(2)^4 \longrightarrow \mathbb{Q}_U(2)^6 \longrightarrow \mathbb{Q}_U(2)^4.$$

It has one surviving copy of $\mathbb{Q}_U(2)$; all other constant terms cancel. The face degrees cancel the displayed shifts on the exceptional ambient and line summands. The resulting Postnikov tower therefore has the three cones in (4.3). Lemma 4.1 identifies the six reduced Artin terms. The two frames are exactly the two extreme classes in the relative cohomology construction, as explained after Definition 3.2. $\square$

4.3. A three-weight category and its primitive kernel. The proof does not assume the existence of a category of all mixed motives over U . We use only the finite triangulated subcategory generated by the objects which have just appeared. Put

$$\mathcal{P}_0 = \langle \mathbb{Q}_U(0) \rangle, \quad \mathcal{P}_1 = \langle \mathbb{Q}_U(1), \chi(1) \rangle, \quad \mathcal{P}_2 = \langle \mathbb{Q}_U(2) \rangle. \quad (4.8)$$

The brackets mean finite direct sums and direct summands. The middle category is the rational Artin category generated by the degree-two cover. If U_p is connected it is the subcategory generated by the trivial and sign representations of its group of order two; if the cover splits, the two displayed generators are isomorphic copies of the trivial object. Either way it is semisimple (by Maschke's theorem in the connected case).

Lemma 4.3 (Three-weight rigidity). *The triangulated subcategory of $\mathrm{DM}(U)_{\mathbb{Q}}$ generated by the three categories in (4.8) has a bounded heart $\mathcal{M}_U^{[0,2]}$, namely*

their extension closure. The two motives of Proposition 4.2 belong to this heart. For rational framed classes one has

$$\ker \overline{\Delta}_{0,2} = \mathrm{Ext}_{\mathcal{M}_U^{[0,2]}}^1(\mathbb{Q}_U(0), \mathbb{Q}_U(2)) = H^1(U, \mathbb{Q}(2)) = 0. \quad (4.9)$$

Proof. We first justify the heart. The semisimple gluing criterion [Gon99, Chapter 5, Section 1] asks for negative morphisms between the pure generators to vanish. Equal-degree groups are direct summands of $H^m(W, \mathbb{Q}(0))$ for $W = U$ or the finite étale cover U_p , and vanish for $m < 0$. Across one increasing Tate degree they are direct summands of $H^m(W, \mathbb{Q}(1))$, which vanish for $m \leq 0$ because $\mathbb{Q}(1) = \mathbf{G}_m[-1]$. Morphisms in the decreasing direction have a negative Tate twist and vanish by effectivity and cancellation: after passing to a finite étale cover to remove an Artin factor, they are motivic cohomology groups of negative codimension, hence zero in the higher-Chow description. The only two-step increasing group is

$$\mathrm{Hom}_{\mathrm{DM}(U)_{\mathbb{Q}}}(\mathbb{Q}_U(0), \mathbb{Q}_U(2)[m]) = H^m(U, \mathbb{Q}(2)).$$

Let $K = \mathbb{Q}(t_1, \dots, t_6)$ be the function field of either rational chart. For $m \leq 1$, Garkusha's birational result and his pure-transcendental invariance theorem give

$$H^m(U, \mathbb{Q}(2)) \cong H^m(K, \mathbb{Q}(2)) \cong H^m(\mathbb{Q}, \mathbb{Q}(2)). \quad (4.10)$$

Here we use Corollaries 3.10, 3.11, 3.13 and Theorem 3.14 of [Gar23]. The groups on the right vanish for $m \leq 0$ by Beilinson–Soulé vanishing over $\mathbb{Q}$. This verifies the remaining negative-Hom condition and constructs the finite heart.

For $m = 1$, motivic cohomology identifies the same group with the weight-two part of algebraic K -theory:

$$H^1(\mathbb{Q}, \mathbb{Q}(2)) = \mathrm{gr}_2^\gamma K_3(\mathbb{Q}) \otimes \mathbb{Q}.$$

Borel's rank theorem says that $K_3(\mathbb{Q})$ has rational rank zero, so this group vanishes.

We also need the strict filtration hypotheses, not merely the existence of the heart. Same-level extension groups are direct summands of $H^1(W, \mathbb{Q}(0))$, which is zero by the higher-Chow description. For $j > i$, both $\mathrm{Hom}(\mathcal{P}_j, \mathcal{P}_i)$ and $\mathrm{Ext}^1(\mathcal{P}_j, \mathcal{P}_i)$ have negative Tate twist and vanish by the preceding effectivity argument. The degree-zero maps between distinct increasing levels were already shown to vanish. Consequently extensions can be reordered uniquely by Tate degree; the heart has the functorial strict filtration required in Lemma 3.4.

That lemma identifies the primitive kernel with $\mathrm{Ext}_{\mathcal{M}_U^{[0,2]}}^1(\mathbb{Q}_U(0), \mathbb{Q}_U(2))$. The heart-to-derived comparison in degree one identifies this with

$$\mathrm{Hom}_{\mathrm{DM}(U)_{\mathbb{Q}}}(\mathbb{Q}_U(0), \mathbb{Q}_U(2)[1]) = H^1(U, \mathbb{Q}(2)),$$

which the Garkusha–Borel calculation has just shown to be zero. This proves every equality in (4.9). $\square$

4.4. The Regge coproduct calculation. For an oriented quadric simplex in projective three-space, Goncharov's middle coproduct has one term for each of its six codimension-two faces. Under the standard weight-one identification with multiplicative units, it is

$$\overline{\Delta}[Q, M] = \sum_e L_e \otimes A_e. \quad (4.11)$$

Here L_e and A_e are the oriented length and angle cross-ratios for the same physical edge e . The order matters because we are working on the projectively dual quadric Q_G . For complementary pairs $\{i, j, k, l\} = \{0, 1, 2, 3\}$, restriction to the coordinate line $M_k \cap M_l$ has binary matrix $G_{\{i,j\},\{i,j\}}$ and gives the angle factor A_{ij} . The quotient by that line has the complementary binary Schur form and gives the length factor L_{kl} . Since the facets i, j meet at the physical edge kl , these are the two factors belonging to that edge. Goncharov's coproduct puts the quotient factor first, hence $L_{kl} \otimes A_{ij}$. Indeed, if S denotes that Schur form on the quotient coordinates k, l , block inversion gives $S^{-1} = (G^{-1})_{\{k,l\},\{k,l\}}$. The normalized dual vectors to the facet normals are the vertices, so

$$\frac{(G^{-1})_{kl}}{\sqrt{(G^{-1})_{kk}(G^{-1})_{ll}}} = \cos \ell_{kl}.$$

Inverting a symmetric binary matrix negates its normalized off-diagonal entry. Therefore the normalized off-diagonal entry of S is $-\cos \ell_{kl}$, exactly the same binary form that produces the length cross-ratio below. The signs are induced by the chosen ruling of Q ; there is no additional alternating edge sign. This is Goncharov's definition of the coproduct, together with its Poincaré-dual description [Gon99, the $S_2(\mathbb{C})$ cross-ratio example, the coproduct definition preceding Theorem 4.5, and Proposition 4.9].

We use the ordered cross-ratio convention

$$r(z_1, z_2, z_3, z_4) = \frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_4)(z_2 - z_3)}.$$

Take affine slope on an oriented real two-plane. The ordered isotropic directions are then $-i, i$, while two real rays meeting at angle ϕ may be written $0, \tan \phi$. A direct calculation gives

$$r(-i, i, 0, \tan \phi) = \frac{1 + i \tan \phi}{1 - i \tan \phi} = e^{2i\phi}.$$

The homogeneous determinant formula for the cross-ratio gives the same identity when one of the slopes is infinite. Swapping the two isotropic directions inverts the answer.

Consequently, if e is a physical edge, ϑ_e its interior dihedral angle, and ℓ_e its length, then with the induced orderings,

$$A_e = e^{2i\vartheta_e}, \quad L_e = e^{2i\ell_e}. \quad (4.12)$$

The two induced orientations are coupled: with the auxiliary induced orientation held fixed, reversing the global ruling reverses exactly one of them, so

every tensor term changes sign, as does the oriented S_4 -generator. In the proof we keep the ruling selected by w fixed throughout.

Let $[\mathbf{m}_G]$ and $[\mathbf{m}_{G^R}]$ denote the two $(0, 2)$ -framed coefficient classes of the normalized relative motives. Their *universal dual Regge defect* is

$$\delta_U^\vee = [\mathbf{m}_G] - [\mathbf{m}_{G^R}] \in A(\mathbb{Q}_U(0), \mathbb{Q}_U(2)).$$

The symbol $\overline{\Delta}_{0,2}$ below is the reduced coproduct in this relative coefficient coalgebra. Its specialization is compared only afterwards with the reduced coproduct in Goncharov's quadric-scissors coalgebra.

Lemma 4.4 (Regge invariance of the middle coproduct). *Let $\delta_U^\vee$ be the difference of the two framed dual quadric motives associated with G and G^R . Then*

$$\overline{\Delta}_{0,2}(\delta_U^\vee) = 0$$

over U , including the possibly nonsplit p -summand. Moreover, for every complex point $s : \text{Spec } \mathbb{C} \rightarrow U$, the corresponding dual quadric-scissors defect satisfies

$$\overline{\Delta}(\delta_s^\vee) = 0 \quad \text{in} \quad (S_2(\mathbb{C}) \otimes S_2(\mathbb{C})) \otimes \mathbb{Q}.$$

Proof. The terms for the two fixed opposite edges agree individually, because Regge fixes both their angle and length. For the four moving terms use the facet-pair order

$$(02, 12, 13, 03)$$

for the angle factors, and attach to them the complementary physical-edge lengths in the order

$$(13, 03, 02, 12).$$

The second list is again a cyclic order around the two fixed opposite edges. Thus Theorem 2.3 applies the same matrix R to the two lists, while the entries with labels 01 and 23 are the two fixed terms. On a finite étale cover choose half-phases

$$A_i = B_i^2, \quad L_i = Y_i^2.$$

Here the cover also includes the roots needed to recover lengths from the angle Gram matrix. To see explicitly that these roots are algebraic, let $C_{ij} = \text{Cof}_{ij}(G)$. If i, j are the endpoints of a geometric edge, the dual-basis construction of the vertices gives, with a sign fixed by the physical branch,

$$\cos \ell_{ij} = \frac{C_{ij}}{\sqrt{C_{ii}C_{jj}}}.$$

Jacobi's complementary-minor identity says

$$C_{ii}C_{jj} - C_{ij}^2 = \det(G) \det G_{\widehat{\{i,j\}}, \widehat{\{i,j\}}},$$

If $\{k, l\}$ is the complementary pair, then

$$\det G_{\widehat{\{i,j\}}, \widehat{\{i,j\}}} = 1 - g_{kl}^2 = - \left(\frac{u_{kl} - u_{kl}^{-1}}{2} \right)^2,$$

where $g_{kl} = -(u_{kl} + u_{kl}^{-1})/2$. Thus the square root of this minor is already supplied by the known angle phase. The identity therefore expresses $\sin \ell_{ij}$, hence $Y_{ij} = e^{i\ell_{ij}}$, using w , square roots of the nonzero principal cofactors, and the phase belonging to the complementary binary angle section. All radicands are units on our chosen open, so adjoining these finitely many roots gives a finite étale cover. If the fixed cover $U_p \rightarrow U$ is connected, include it as one more factor in this fibre product.

Choose the connected component V of the resulting cover which contains the physical choices $B_i = e^{i\theta_i}$ and $Y_i = e^{i\ell_i}$. The map $V \rightarrow U$ is surjective: a connected component of a finite étale cover has open-and-closed image, and U is connected. When U_p is connected, the same argument shows that the induced map $V \rightarrow U_p$ is surjective. There is a real open neighbourhood Ω of the given tetrahedron on which both Gram matrices remain positive definite. Its phase image is a nonempty maximal totally real open subset of the six-dimensional complex manifold $U(\mathbb{C})$: algebraically, the full phase orientation cover maps finitely and dominantly to U , forgetting only the choice between p and p^{-1} , and the six real angles are local coordinates on the physical chamber. A holomorphic function which vanishes there is zero by the one-variable identity theorem applied successively in local analytic coordinates; hence the image is Zariski dense. Choose the square-root branches analytically over a smaller simply connected part of Ω . Their lift $\tilde{\Omega} \subset V(\mathbb{C})$ is a nonempty maximal totally real open set in the complex manifold $V(\mathbb{C})$. A regular function on the integral variety V which vanishes on $\tilde{\Omega}$ is holomorphic in local coordinates and hence vanishes identically by the same successive identity theorem. Thus $\tilde{\Omega}$ is Zariski dense in the chosen component V .

Let $R = \frac{1}{2}J_4 - I_4$ be the Regge matrix and put $H = 2R = J_4 - 2I_4$. The angle and length parts of the Regge theorem give, first on the real spherical chamber,

$$A_i^R = \prod_j B_j^{H_{ij}}, \quad L_i^R = \prod_j Y_j^{H_{ij}}. \quad (4.13)$$

Both sides are algebraic functions on V . Since $\tilde{\Omega}$ is Zariski dense there, the identities hold algebraically over its function field. We write a multiplicative group additively inside a tensor product; in particular

$$[u^m] \otimes [v^n] = mn [u] \otimes [v].$$

Since $H^t H = 4I_4$, this convention gives

$$\begin{aligned} \sum_i L_i^R \otimes A_i^R &= \sum_{i,j,k} H_{ik} H_{ij} Y_k \otimes B_j \\ &= 4 \sum_j Y_j \otimes B_j = \sum_j L_j \otimes A_j. \end{aligned}$$

Changing a chosen square root contributes only two-torsion, which disappears over $\mathbb{Q}$.

All functions in this tensor identity are regular units after the shrinking made at the start of this section. For an integral scheme V , the map $\mathcal{O}(V)^* \hookrightarrow K(V)^*$ is injective, and remains so after tensoring with $\mathbb{Q}$; the same is true of its tensor square over $\mathbb{Q}$. Consequently the identity proved in the function field is already an identity of the relative weight-one coefficient objects on the cover.

We finally descend the equality. If $V \rightarrow U$ is a finite étale cover with structure map f , then on units

$$N_{V/U} \circ f^* = [\deg(V/U)].$$

Thus f^* on rationalized units is injective, and its tensor square is injective as well. This descends the equality formed by the five split channels (the four moving terms and the fixed q -term). If U_p is split, the p -channel is a sixth ordinary-unit channel and the same argument with $V \rightarrow U$ finishes the descent. Suppose instead that U_p is connected. The surjective finite étale map $V \rightarrow U_p$ has an injective pullback on rationalized units by its norm, so the fixed p -term first descends to U_p . For the deck involution σ of $\pi : U_p \rightarrow U$, put

$$e_- = \frac{1 - \sigma}{2}, \quad \chi = e_- M_U(U_p).$$

Using $\pi^* \chi \cong \mathbb{Q}_{U_p}$, pullback on the two half-frame spaces is

$$\begin{aligned} \pi^* : A_U(\mathbb{Q}(0), \chi(1)) &\longrightarrow A_{U_p}(\mathbb{Q}(0), \mathbb{Q}(1)), \\ \pi^* : A_U(\chi(1), \mathbb{Q}(2)) &\longrightarrow A_{U_p}(\mathbb{Q}(1), \mathbb{Q}(2)). \end{aligned}$$

On either space, the normalized twisted trace—the χ -isotypic projection of transfer, divided by two—is a left inverse: trace after the adjunction unit is multiplication by the degree, namely two, on the sign summand. The tensor product of these two left inverses proves injectivity on the complete p -term. That term is identical for the two mates on U_p , since Regge fixes both the p -angle and its complementary length. Its difference therefore vanishes already over U . Equation (4.11) now proves the relative motivic assertion.

The calculation has taken place in the rationalized multiplicative unit groups themselves, before applying any period or regulator. Goncharov's cross-ratio isomorphism identifies $S_2(\mathbb{C}) \otimes \mathbb{Q}$ with $\mathbb{C}^* \otimes \mathbb{Q}$, and (4.11) is also the coproduct formula in his quadric-scissors Hopf algebra. Pulling the same unit identity to any complex point therefore proves $\bar{\Delta}(\delta_s^\vee) = 0$ in the stated scissors target. $\square$

Theorem 4.5 (Vanishing of the universal framed defect). *The universal framed coefficient class is zero:*

$$\delta_U^\vee = 0 \quad \text{in} \quad A(\mathbb{Q}_U(0), \mathbb{Q}_U(2))$$

for the restricted three-weight heart $\mathcal{M}_U^{[0,2]}$.

Proof. Proposition 4.2 places the two framed motives, and hence their difference, in the heart of Lemma 4.3. Lemma 4.4 says that their difference is primitive. The primitive kernel is zero by (4.9). $\square$

This theorem is an identity over the parameter space U , not a density or continuity statement. We next pull it back to the individual complex point determined by the original tetrahedron and identify exactly what it says about its scissors class.

5. SPECIALIZATION AND THE RETURN TO SCISSORS CONGRUENCE

Theorem 4.5 is a relative identity over the algebraic parameter space U . We now explain, without suppressing the comparison maps, why its fibre at the original tetrahedron proves Theorem 1.1.

5.1. The two weight-two injections. Let $S_4(\mathbb{C})$ denote Goncharov's abelian group of oriented smooth quadrics in $\mathbf{P}^3$ equipped with four ordered hyperplane faces, modulo the relations recalled in Section 3. Put $S_j(\mathbb{C})_{\mathbb{Q}} = S_j(\mathbb{C}) \otimes_{\mathbf{Z}} \mathbb{Q}$. In weight one the cross-ratio isomorphism identifies $S_2(\mathbb{C})_{\mathbb{Q}}$ with $\mathbb{C}^* \otimes_{\mathbf{Z}} \mathbb{Q}$. The weight-two part of the normalized cobar complex begins, in cohomological degrees one and two, as

$$S_4(\mathbb{C})_{\mathbb{Q}} \xrightarrow{\bar{\Delta}} S_2(\mathbb{C})_{\mathbb{Q}} \otimes_{\mathbb{Q}} S_2(\mathbb{C})_{\mathbb{Q}}.$$

There is no degree-zero term. Consequently its degree-one cohomology is

$$H^1(S^{\bullet}(2))_{\mathbb{Q}} = \ker \bar{\Delta}. \quad (5.1)$$

We use the following two theorems of Goncharov.

Theorem 5.1 (Goncharov's weight-two comparison). *There is a canonical map*

$$c_G : H^1(S^{\bullet}(2))_{\mathbb{Q}} \longrightarrow \mathrm{gr}_2^{\gamma} K_3(\mathbb{C}) \otimes \mathbb{Q}, \quad (5.2)$$

obtained from the framed relative motive of a quadric simplex. It is injective.

Proof. This is the case $n = 2, i = 1$ of [Gon99, Theorem 4.13, equation (4.7)]. The theorem proves in weight two both the existence and injectivity asserted in Conjecture 4.11 of that paper. $\square$

Let

$$\tilde{\mathcal{P}}(S^3) = \mathrm{coker}(\Sigma : \hat{\mathcal{P}}(S^2) \longrightarrow \hat{\mathcal{P}}(S^3))$$

be the reduced spherical scissors group; Σ is spherical suspension. If P lies in an equatorial $S^2 \subset S^3$, then ΣP is the double cone obtained by joining P to the two antipodal poles. Cutting and isometries make this construction an additive map on scissors groups.

Lemma 5.2 (Comparison of the two spherical presentations). *Let $\mathcal{P}_G(S^3)$ be Goncharov's group generated by oriented, ordered spherical tetrahedra with his degeneracy, isometry, permutation, and five-term additivity relations. There is a canonical isomorphism*

$$\tilde{\mathcal{P}}(S^3) \xrightarrow{\sim} \mathcal{P}_G(S^3).$$

Proof. We spell out the comparison because the unreduced and reduced spherical groups must not be confused. Let $C_k(S^3)$ be the free abelian group on ordered $(k + 1)$ -tuples of points of S^3 , with boundary

$$\partial(x_0, \dots, x_k) = \sum_{i=0}^k (-1)^i (x_0, \dots, \widehat{x_i}, \dots, x_k).$$

Let $D_* \subset C_*(S^3)$ be the subcomplex generated by tuples lying in a proper great subsphere, and put

$$\text{St}(S^3) = H_3(C_*(S^3)/D_*).$$

A proper ordered four-tuple is a cycle in this quotient because each of its three-point faces lies in a great S^2 ; a degenerate tuple is zero; and the boundary of an ordered five-tuple is exactly the alternating five-term relation. Recording an ambient orientation and alternating in the four entries gives Goncharov's permutation rule. Taking $H_0(SO(4), \text{St}(S^3))$, that is, coinvariants *after* forming this homology group, imposes the oriented-isometry relation. Allowing $O(4)$ while transporting the orientation gives the same quotient by Gerling's theorem. Thus Goncharov's presentation is $H_0(SO(4), \text{St}(S^3))$.

The proof of Dupont's Theorem 7.14 identifies this group canonically with

$$\widehat{\mathcal{P}}(S^3)/\Sigma\widehat{\mathcal{P}}(S^2) = \widetilde{\mathcal{P}}(S^3).$$

The theorem further identifies it with the degree-three homology of the full coinvariant configuration complex; this further description is not needed here. Under the first isomorphism, a proper ordered tuple maps to its oriented geometric simplex. This proves the claim [Dup01, Theorem 7.14] [Gon99, Chapter 2, Section 1]. $\square$

Theorem 5.3 (Goncharov's spherical map). *Let*

$$S_{4,\mathbb{Q}}^{\text{sph}} = (S_4(\mathbb{C}) \otimes_{\mathbf{Z}} \mathbf{Z}(2)_{\text{par}})^+ \otimes_{\mathbf{Z}} \mathbb{Q}, \quad \mathbf{Z}(2)_{\text{par}} = (2\pi i)^2 \mathbf{Z},$$

where complex conjugation acts on both factors and $^+$ denotes the invariant part. Then complexification of a spherical simplex gives a canonical injective map

$$\psi_S(2) : \widetilde{\mathcal{P}}(S^3) \otimes \mathbb{Q} \hookrightarrow S_{4,\mathbb{Q}}^{\text{sph}}. \quad (5.3)$$

It takes the spherical Dehn differential to the quadric coproduct. For a tetrahedron with angle Gram matrix G , its value is represented by the oriented coordinate quadric simplex $Q_{G^{-1}}$ with its four coordinate hyperplanes.

Proof. Use Lemma 5.2 to identify the source with Goncharov's group. The asserted map and its compatibility with the coproduct are the spherical parts of [Gon99, Theorem 4.14]; its injectivity is stated immediately after the proof of that theorem. To see the last description concretely, let $n_0, \dots, n_3$ be the four outward facet normals. Their Gram matrix is G . Write a vector in the basis dual to the n_i , or equivalently use the four facet linear forms $v \mapsto \langle n_i, v \rangle$ as coordinates. In those coordinates the ambient spherical quadratic form

has matrix G^{-1} , and the four facets are the coordinate hyperplanes. The determinant root selects the same family of maximal isotropic planes used to orient Goncharov's generator. We choose the ambient ordered orientations of the two real simplices so that these roots are the common lift $(G, w) \mapsto (G^R, w)$ of (2.25). Gerling's theorem permits this orientation choice without changing the ordinary or reduced spherical scissors classes. $\square$

The line $\mathbf{Z}(2)_{\text{par}}$ only records conjugation parity; it is not the motivic Tate object $\mathbb{Q}(2)$ used in $H^1(-, \mathbb{Q}(2))$. After tensoring with $\mathbb{Q}$ it is one-dimensional and may be forgotten when testing whether a class vanishes. Goncharov's projective-duality map is an involutive linear automorphism of $S_4(\mathbb{C})_{\mathbb{Q}}$; it commutes with conjugation and hence preserves the spherical invariant part. Thus a class vanishes if and only if its dual class, represented in our coordinates by Q_G , vanishes.

5.2. Why the relative motive is the coefficient in c_G . There is one comparison which must be made literally. Knowing that some motivic realization of a quadric class vanishes would not suffice unless it were the realization occurring in (5.2).

Lemma 5.4 (Fibre comparison). *Let $s : \text{Spec } \mathbb{C} \rightarrow U$ be a complex point. Let $\delta_s^\vee$ be the primitive difference of the two dual quadric-scissors generators, regarded as an element of $\ker \bar{\Delta} \subset S_4(\mathbb{C})_{\mathbb{Q}}$. The endpoint class obtained by pulling back the relative framed defect $\delta_U^\vee$ is exactly*

$$c_G(\delta_s^\vee) \in H^1(\mathbb{C}, \mathbb{Q}(2)) = \text{gr}_2^\gamma K_3(\mathbb{C}) \otimes \mathbb{Q}.$$

Proof. For any smooth base T , write

$$\Phi_T(Q, \alpha; M) = [D\tilde{m}_T(Q, M)(2); v_\alpha, f_M]$$

for the framed coefficient of an oriented quadric simplex. Here $\tilde{m}_T(Q, M)$ is the alternating face complex of Definition 3.1; the ruling α fixes the extreme frame v_α , and the ordered oriented hyperplanes fix the other frame f_M . Interchanging the ruling or applying an odd permutation changes exactly one frame, so this has Goncharov's sign conventions.

This is not a new realization of his generator. Chapter 5, Section 2 of [Gon99] defines

$$m(Q, L) = \underline{\text{Hom}}(\tilde{m}(Q, L), \mathbb{Q})$$

from the same four-term face complex, with the zero-dimensional faces in degree 0 and $\mathbf{P}^3 \setminus Q$ in degree 3. Hence no cohomological shift occurs. In our convention $\mathbb{Q}(1) = \mathbf{G}_m[-1]$, its raw extreme Tate factors are $\mathbb{Q}(0)$ and $\mathbb{Q}(-2)$; tensoring by $\mathbb{Q}(2)$ rewrites the coefficient as an extension of $\mathbb{Q}(0)$ by $\mathbb{Q}(2)$. Thus Goncharov's coefficient, in our normalization, is exactly $\Phi_T = \mathbf{m}_T$. This is also Brown's normalization [Bro13, Definition 5.8]. The modern category $\text{DM}(T)_{\mathbb{Q}}$ represents the same bounded complex of smooth strata, its internal dual, and its two geometric frame maps.

Every face term is constructible and dualizable. Exact symmetric monoidal pullback therefore gives a canonical framed isomorphism

$$s^*\Phi_U(Q, \alpha; M) \simeq \Phi_{\mathbb{C}}(Q_s, \alpha_s; M_s). \quad (5.4)$$

Indeed, pullback commutes term by term with the incidence maps, internal duality, the Tate twist, and the frames. It also preserves the explicit face filtration and its endpoint connecting morphism. No motivic t -structure over $\mathbb{C}$, and no t -exactness assertion, is needed.

A primitive coefficient determines an endpoint triangle

$$\mathbb{Q}_U(2) \longrightarrow E \longrightarrow \mathbb{Q}_U(0) \xrightarrow{\kappa_U} \mathbb{Q}_U(2)[1]. \quad (5.5)$$

Equations (4.6)–(4.7) of [Gon99] first send a quadric-simplex generator to $\Phi_{\mathbb{C}}$, then apply degree-one normalized cobar cohomology. In weight two the resulting map is

$$\begin{aligned} H^1(\Phi_{\mathbb{C}}) : H^1(S^\bullet(2))_{\mathbb{Q}} &\longrightarrow \mathrm{Hom}_{\mathrm{DM}(\mathbb{C})_{\mathbb{Q}}}(\mathbb{Q}(0), \mathbb{Q}(2)[1]) \\ &\cong \mathrm{gr}_2^\gamma K_3(\mathbb{C}) \otimes \mathbb{Q}, \end{aligned} \quad (5.6)$$

and this is the map c_G of Theorem 5.1. Because the quadric cobar complex starts in degree 1, a primitive element of $S_4(\mathbb{C})_{\mathbb{Q}}$ is literally a degree-one cocycle. On such a cocycle (5.6) is its endpoint connecting morphism: the reduced coproduct is exactly the obstruction contributed by the middle Tate degree. Here $H^1(\Phi_{\mathbb{C}})$ is shorthand for this direct operation on primitive degree-one cocycles; we are not asserting a global abelian mixed-Tate heart over $\mathbb{C}$.

Apply this construction to the additive difference $\delta_U^\vee$. Equation (5.4) identifies its fibre with the coefficient of $\delta_s^\vee$, and naturality of the endpoint triangle gives

$$s^*\kappa_U(\delta_U^\vee) = \kappa_{\mathbb{C}}(\Phi_{\mathbb{C}}(\delta_s^\vee)) = H^1(\Phi_{\mathbb{C}})(\delta_s^\vee) = c_G(\delta_s^\vee).$$

This is an equality in motivic cohomology before any regulator is applied.

Finally, two different dualities occur in the notation. The superscript $\vee$ is projective duality, the antipode taking $Q_{G^{-1}}$ to Q_G and transporting its ruling. The D inside Φ_T is the internal cohomological dual of the resulting face complex. Thus no second unrecorded projective dual or sign is present. Theorem 4.13 proves the existence and injectivity of (5.6) in weight two. Goncharov's stronger Conjecture 4.15 asks for an explicit chain map from the entire quadric complex to a Bloch complex; neither that chain map nor any Bloch-symbol formula is used here. $\square$

5.3. Specializing to the physical tetrahedron. Let T, T^R be the pair in Theorem 1.1. By Proposition 2.8, its physical angle phases and positive determinant root define a point

$$s_T : \mathrm{Spec} \mathbb{C} \longrightarrow U$$

on one of the two rational charts. Put

$$z = [T] - [T^R] \in \widehat{\mathcal{P}}(S^3), \quad \bar{z} \in \widetilde{\mathcal{P}}(S^3).$$

The spherical map sends $\bar{z} \otimes 1$ to $\delta_{s_T} \otimes (2\pi i)^2$. Forgetting this nonzero one-dimensional parity factor, we regard δ_{s_T} as an element of $S_4(\mathbb{C})_{\mathbb{Q}}$. Projective duality sends it to $\delta_{s_T}^{\vee}$, the fibre of the relative defect considered in Section 4. Lemma 4.4, in its quadric-scissors form, makes it primitive.

Theorem 4.5 says that $\delta_U^{\vee} = 0$ as a framed class. Therefore its endpoint morphism is zero, and Lemma 5.4 gives

$$c_G(\delta_{s_T}^{\vee}) = 0.$$

The injectivity of c_G in Theorem 5.1 implies $\delta_{s_T}^{\vee} = 0$ in $S_4(\mathbb{C})_{\mathbb{Q}}$, and projective duality then gives $\delta_{s_T} = 0$ there. Injectivity of $\psi_S(2)$ yields

$$\bar{z} = 0 \quad \text{in } \tilde{\mathcal{P}}(S^3) \otimes \mathbb{Q}. \quad (5.7)$$

This completes the motivic part of the proof. The remaining paragraph is ordinary spherical scissors geometry.

5.4. Removing rationalization and suspension. We record the three standard spherical scissors facts used here [Dup01, Example 7.8 and Corollary 9.19(b)]:

- (S1) $\hat{\mathcal{P}}(S^3)$ is uniquely divisible, hence is a $\mathbb{Q}$ -vector space.
- (S2) Area is an isomorphism

$$\text{Area} : \hat{\mathcal{P}}(S^2) \xrightarrow{\sim} \mathbb{R}.$$

- (S3) Equality in the spherical scissors group can be converted into a finite equidecomposition by Zylev cancellation; Gerling's theorem permits the motions to be taken in $SO(4)$ [Dup01, after Definition 1.6 and Theorem 2.2].

Since $\hat{\mathcal{P}}(S^2) \cong \mathbb{R}$ and $\hat{\mathcal{P}}(S^3)$ are $\mathbb{Q}$ -vector spaces, the image of suspension is a $\mathbb{Q}$ -subspace. Its cokernel $\tilde{\mathcal{P}}(S^3)$ is therefore also a $\mathbb{Q}$ -vector space. Consequently (5.7) already says $\bar{z} = 0$ before tensoring with $\mathbb{Q}$. By the definition of the reduced group, there is a class $Q \in \hat{\mathcal{P}}(S^2)$ with

$$z = \Sigma Q. \quad (5.8)$$

If a spherical polygon of area A is suspended from the two poles of S^3 , the warped-product volume element is $\cos^2(t) dt dA$, where $-\pi/2 \leq t \leq \pi/2$. Therefore

$$\text{Vol}(\Sigma Q) = \left(\int_{-\pi/2}^{\pi/2} \cos^2 t dt \right) \text{Area}(Q) = \frac{\pi}{2} \text{Area}(Q). \quad (5.9)$$

Both sides are additive under cutting, so (5.9) holds for every class $Q \in \hat{\mathcal{P}}(S^2)$, not only for a single polygon. Regge mates have equal volume by Theorem 2.3. Applying volume to (5.8) gives $\text{Area}(Q) = 0$. The area classification forces $Q = 0$, so $z = 0$ in $\hat{\mathcal{P}}(S^3)$.

Zylev cancellation and Gerling's theorem now turn this equality into a finite equidecomposition. This proves Theorem 1.1.

REFERENCES

- [AI19] Arseniy Akopyan and Ivan Izmistiev. The regge symmetry, confocal conics, and the schläfli formula. *Bulletin of the London Mathematical Society*, 51(5):765–775, 2019.
- [BGSV90] A. A. Beilinson, A. B. Goncharov, V. V. Schechtman, and A. N. Varchenko. Aomoto dilogarithms, mixed Hodge structures and motivic cohomology of pairs of triangles on the plane. In *The Grothendieck Festschrift, Vol. I*, volume 86 of *Progress in Mathematics*, pages 135–172. Birkhäuser, Boston, 1990.
- [Bor77] Armand Borel. Cohomologie de SL_n et valeurs de fonctions zêta aux points entiers. *Annali della Scuola Normale Superiore di Pisa, Classe di Scienze*, 4(4):613–636, 1977.
- [Bro13] Francis C. S. Brown. Dedekind zeta motives for totally real fields. *Inventiones Mathematicae*, 194(2):257–311, 2013.
- [Dup01] Johan L. Dupont. *Scissors Congruences, Group Homology and Characteristic Classes*, volume 1 of *Nankai Tracts in Mathematics*. World Scientific, River Edge, NJ, 2001.
- [Gar23] Grigory Garkusha. Semilocal Milnor K -theory. *International Mathematics Research Notices*, 2023(24):22069–22095, 2023.
- [Gon99] Alexander B. Goncharov. Volumes of hyperbolic manifolds and mixed Tate motives. *Journal of the American Mathematical Society*, 12(2):569–618, 1999.